



HK IMO Training 2023-2024 Problem Set 2 Suggested Solutions



Problem 5 (Pan-African MO 2016 Problem 2 modified). There are n cards in a single pile. In each move, we can choose a pile with at least 3 cards, remove 1 card from this pile, and divide the remaining cards in this pile into two nonempty piles (not necessarily with the same number of cards). Find all positive integers n such that it is possible to make all piles have exactly 3 cards after some moves.

Solution. Answer: All positive integers n of the form $4k - 1$.

Firstly, suppose there are k piles at the end. This means there are $3k$ cards in total at the end. Note that the number of piles is increased by 1 in each move. Therefore, we have carried out $k - 1$ moves, and so $k - 1$ cards are removed. The original pile must contain exactly $3k + (k - 1) = 4k - 1$ cards.

Secondly, we show that all positive integers $n = 4k - 1$ are admissible. For example, in each move we can divide the pile with the most number of cards into two piles, including one pile with exactly 3 cards. It is easy to see that the pile with the most number of cards contains 4 cards fewer after each move. Eventually, this pile contains exactly 3 cards as $4k - 1 \equiv 3 \pmod{4}$. At this stage all piles have exactly 3 cards. $\square$

Problem 6 (IRN-TWN FdMO 2022 Problem 1). Let $n \geq 2$ be an integer, and let x and y be real numbers. Prove that $x - y$ is an integer divisible by n if and only if

$$\lfloor kx \rfloor \equiv \lfloor ky \rfloor \pmod{n}$$

for every positive integer k .

Solution. If $x - y$ is an integer divisible by n , let $x - y = mn$ where $m \in \mathbb{Z}$. Then

$$\lfloor kx \rfloor - \lfloor ky \rfloor = \lfloor ky + kmn \rfloor - \lfloor ky \rfloor = \lfloor ky \rfloor + kmn - \lfloor ky \rfloor = kmn \equiv 0 \pmod{n}$$

for every positive integer k .

Conversely, if

$$\lfloor kx \rfloor \equiv \lfloor ky \rfloor \pmod{n}$$

for every positive integer k , let $x = a + \alpha$ and $y = b + \beta$ where $a = \lfloor x \rfloor$ and $b = \lfloor y \rfloor$. Note that $\lfloor kx \rfloor = ka + \lfloor k\alpha \rfloor$ and $\lfloor ky \rfloor = kb + \lfloor k\beta \rfloor$. Putting $k = 1$, we obtain

$$a + \lfloor \alpha \rfloor \equiv b + \lfloor \beta \rfloor \pmod{n}.$$

This implies $a \equiv b \pmod{n}$ as $0 \leq \alpha, \beta < 1$. Now, for arbitrary k , the condition becomes

$$ka + \lfloor k\alpha \rfloor \equiv kb + \lfloor k\beta \rfloor \pmod{n},$$

which means

$$\lfloor k\alpha \rfloor \equiv \lfloor k\beta \rfloor \pmod{n}.$$

We are done if $\alpha = \beta$, since this implies $x - y = a - b \equiv 0 \pmod{n}$. Suppose $\alpha \neq \beta$. WLOG assume $\alpha > \beta$. Let t be the smallest positive integer such that $\frac{\ell}{2^t} \in (\beta, \alpha)$ for some integer ℓ . Note that ℓ is uniquely determined. Otherwise, there exists ℓ such that $\frac{\ell}{2^t}, \frac{\ell+1}{2^t} \in (\beta, \alpha)$, but one of $\frac{\ell}{2^t}$ and $\frac{\ell+1}{2^t}$ has an even numerator, which means it can be simplified to fraction with a smaller denominator (hence t is not the smallest).

As there is a unique integer ℓ satisfying $\alpha > \frac{\ell}{2^t} > \beta$, there is a unique integer ℓ satisfying $2^t \alpha > \ell > 2^t \beta$. Therefore, we have $\lfloor 2^t \alpha \rfloor - \lfloor 2^t \beta \rfloor = 1$, which is not divisible by n . This contradicts the assumption with $k = 2^t$. Therefore, we must have $\alpha = \beta$. $\square$

Problem 7 (IRNMO 2016 2nd Problem 1). Let a, b, c be real numbers satisfying $0 < a \leq b \leq c$. Prove that

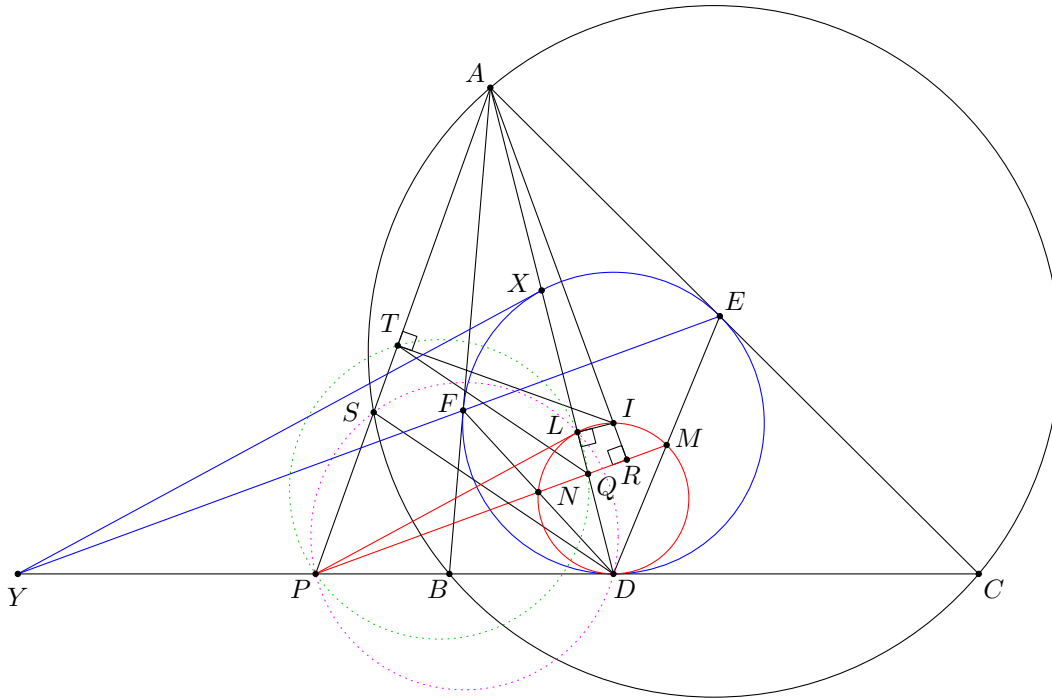
$$\frac{a+b+c}{3} - \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \geq \frac{(c-a)^2}{6c}.$$

Solution. Firstly, note that $\frac{(c-a)^2}{6c} \leq \frac{(c-a)^2}{3(a+c)}$ because $a \leq c$. Secondly, note that

$$\begin{aligned} & \frac{a+b+c}{3} - \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \geq \frac{(c-a)^2}{3(a+c)} \\ \Leftrightarrow & \frac{a+b+c}{3} - \frac{3abc}{ab+bc+ca} \geq \frac{(c-a)^2}{3(a+c)} \\ \Leftrightarrow & \frac{a^2b + a^2c + b^2a + b^2c + c^2a + c^2b - 6abc}{3(ab+bc+ca)} \geq \frac{(c-a)^2}{3(a+c)} \\ \Leftrightarrow & (a^2b + a^2c + b^2a + b^2c + c^2a + c^2b - 6abc)(a+c) \geq (c-a)^2(ab+bc+ca) \\ \Leftrightarrow & (a+c)^2b^2 - 4ac(a+c)b + 4a^2c^2 \geq 0 \\ \Leftrightarrow & ((a+c)b - 2ac)^2 \geq 0. \end{aligned}$$

This is obviously true. Equality holds when $a = b = c$. □

Problem 8 (IRN-TWN FdMO 2022 Problem 3). Let I be the incentre of a scalene $\triangle ABC$. The incircle of $\triangle ABC$ touches BC, CA, AB at D, E, F respectively. Let M and N be the midpoints of DE and DF respectively. The line MN intersects the lines BC and AD at P and Q respectively. Suppose AP meets the circumcircle of $\triangle ABC$ again at S . Let T be the foot of perpendicular from I to AP . Prove that SD and TQ are parallel.



Solution. Let L be the foot of perpendicular from I to AD , and let R be the intersection point of the lines AI and MN . Since $MN \parallel EF$, we have $MN \perp AI$. Therefore, I, L, N, R are concyclic. Also, A, T, L, I are concyclic. It follows from the property of the Miquel point that P, Q, L, T are concyclic (or from $\angle ATL = \angle RIL = \angle PNL$).

Next, let AD meet the incircle again at X . It is well-known that DX is a symmedian of $\triangle DEF$. Therefore, the line EF and the tangents at D and X to the incircle are concurrent at some point Y . Note that L is the midpoint of DX since $IL \perp DX$. Thus, there is a homothety at centre D that maps E, F, X, Y to M, N, L, P respectively. In particular, we know that P is the midpoint of DY and $PD = PL$.

It is well-known that $(Y, D; B, C) = -1$. As P is the midpoint of DY , we have $PD^2 = PB \times PC = PS \times PA$. This shows PD is tangent to (ASD) , and hence $\angle PSD = \angle ADP = \angle LDP = \angle PLD = \angle PLQ = \angle PTQ$. This proves $SD \parallel TQ$. □